

# Atmospheric Radiation Measurement User Facility



ARM Data Quality Report (DQR) Table

| Datastream           | DQR        | Quality   | Subject                                                                       | Start Date          | End Date            |
|----------------------|------------|-----------|-------------------------------------------------------------------------------|---------------------|---------------------|
| bnfaerisummaryS40.b1 | D250909.1  | Missing   | Missing Data                                                                  | 2025-09-05T19:39:49 | 2025-09-08T18:34:58 |
| bnfaerisummaryS40.b1 | D260217.2  | Missing   | Missing Data                                                                  | 2026-01-08T04:50:24 | 2026-01-14T17:48:59 |
| bnfaerisummaryS40.b1 | D260217.1  | Missing   | Missing Data                                                                  | 2026-01-31T03:46:39 | 2026-02-02T15:11:25 |
| bnfaerisummaryS40.b1 | D241226.2  | Missing   | Mirror Controller/Motor Communication Issues                                  | 2024-12-18T09:53:00 | 2024-12-20T21:59:28 |
| bnfaerisummaryS40.b1 | D241226.1  | Incorrect | Housekeeping Meter Malfunction                                                | 2024-12-18T00:00:00 | 2024-12-18T09:53:00 |
| bnfdlfptS40.b1       | D250208.1  | Missing   | No Data                                                                       | 2024-10-28T15:15:32 | 2024-10-30T15:30:04 |
| bnfdlfptS40.b1       | D250210.2  | Missing   | No Data                                                                       | 2024-11-30T12:45:32 | 2024-12-02T17:00:19 |
| bnfdlfptS40.b1       | D250728.8  | Missing   | No Data                                                                       | 2025-05-06T20:17:01 | 2025-05-13T16:15:59 |
| bnfecorsfS40.b1      | D250404.2  | Missing   | No Data                                                                       | 2025-03-13T14:00:01 | 2025-04-03T15:59:59 |
| bnfgndrad60sS40.b1   | D250825.1  | Incorrect | Incorrect Data                                                                | 2025-05-18T03:59:00 | 2025-08-19T16:00:00 |
| bnfirtS40.b1         | D250307.8  | Missing   | Missing Data                                                                  | 2025-01-15T16:31:01 | 2025-01-22T14:33:59 |
| bnfirtS40.b1         | D250627.10 | Missing   | No Data                                                                       | 2025-05-18T04:20:01 | 2025-05-19T14:05:59 |
| bnfirtS40.b1         | D241011.3  | Missing   | No Data                                                                       | 2024-10-01T00:00:00 | 2024-10-04T22:59:59 |
| bnfmfrsr7nchS40.b1   | D250714.19 | Missing   | Missing Data                                                                  | 2025-04-06T09:14:01 | 2025-04-07T15:14:40 |
| bnfmfrsr7nchS40.b1   | D250512.6  | Missing   | Missing Data                                                                  | 2025-04-06T09:14:01 | 2025-04-07T15:17:20 |
| bnfmfrsr7nchS40.b1   | D250611.5  | Incorrect | Incorrect Data                                                                | 2024-08-26T00:00:00 | 2025-05-22T17:46:00 |
| bnfmfrsr7nchS40.b1   | D251024.11 | Incorrect | Incorrect Data                                                                | 2025-08-31T03:04:20 | 2025-10-09T05:59:40 |
| bnfskyrad60sS40.b1   | D250310.1  | Incorrect | Incorrect Data                                                                | 2025-03-06T03:11:00 | 2025-03-06T12:12:00 |
| bnfskyrad60sS40.b1   | D260102.1  | Incorrect | Incorrect Data                                                                | 2025-12-31T18:31:00 | 2025-12-31T19:02:00 |
| bnfskyrad60sS40.b1   | D250424.1  | Incorrect | Incorrect Data                                                                | 2025-04-06T05:41:00 | 2025-04-23T15:54:00 |
| bnfskyrad60sS40.b1   | D251210.2  | Incorrect | Incorrect Data                                                                | 2025-12-09T14:30:00 | 2025-12-09T16:57:00 |
| bnfskyrad60sS40.b1   | D251014.1  | Incorrect | Incorrect Data                                                                | 2025-08-28T00:00:00 | 2025-09-05T19:30:00 |
| bnfskyrad60sS40.b1   | D260413.1  | Suspect   | Questionable Data                                                             | 2026-04-05T11:34:00 | 2026-04-05T23:23:59 |
| bnfskyrad60sS40.b1   | D250108.7  | Suspect   | Excessive (and Increasing) down_long_hemisp1 and down_long_hemips2 Divergence | 2024-12-26T22:00:00 | 2025-01-07T20:56:00 |
| bnfskyrad60sS40.b1   | D260107.4  | Suspect   | Questionable Data                                                             | 2025-12-25T23:00:00 | 2025-12-26T07:00:00 |

## ARM Data Object Identifier (DOI) Table

| Instrument | DOI                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AERI       | Gero, J., Garcia, R., Hackel, D., Ermold, B., & Gaustad, K. Atmospheric Emitted Radiance Interferometer (AERICH1), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1989299">https://doi.org/10.5439/1989299</a>                                                                       |
| DL         | Newsom, R., Shi, Y., & Krishnamurthy, R. Doppler Lidar (DLCAL1), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1025183">https://doi.org/10.5439/1025183</a>                                                                                                                         |
| MWR3C      | Cadeddu, M., Gibler, G., Koontz, A., & Tuftedal, M. Microwave Radiometer, 3 Channel (MWR3C), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1025248">https://doi.org/10.5439/1025248</a>                                                                                             |
| ECOR       | Sullivan, R., Cook, D., Shi, Y., Keeler, E., & Pal, S. Eddy Correlation Flux Measurement System (ECORSF), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, Supplemental facility for STAMP2 near Tower Site (S13). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1494128">https://doi.org/10.5439/1494128</a>                                               |
| GNDRAD     | Sengupta, M., Habte, A., Andreas, A., Reda, I., Jaker, S., Xie, Y., Yang, J., Gotseff, P., Kutchenreiter, M., & Shi, Y. Ground Radiometers on Stand for Upwelling Radiation (GNDRAD60S), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1377837">https://doi.org/10.5439/1377837</a> |
| IRT        | Shi, Y., Howie, J., Goldberger, L., & Morris, V. Infrared Thermometer (IRT), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1329238">https://doi.org/10.5439/1329238</a>                                                                                                             |
| MET        | Kyrouac, J., Shi, Y., & Tuftedal, M. Surface Meteorological Instrumentation (MET), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1786358">https://doi.org/10.5439/1786358</a>                                                                                                       |
| MFRSR      | Hodges, G., Sheffer, B., Herrera, C., & Ermold, B. Multifilter Rotating Shadowband Radiometer (MFRSR7NCH), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1429369">https://doi.org/10.5439/1429369</a>                                                                               |
| SEBS       | Sullivan, R., Keeler, E., Pal, S., & Kyrouac, J. Surface Energy Balance System (SEBS), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Tower Site) (S10). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1984921">https://doi.org/10.5439/1984921</a>                                                                                                 |
| SKYRAD     | Sengupta, M., Habte, A., Andreas, A., Reda, I., Jaker, S., Xie, Y., Yang, J., Gotseff, P., Kutchenreiter, M., & Shi, Y. Sky Radiometers on Stand for Downwelling Radiation (SKYRAD60S), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1377836">https://doi.org/10.5439/1377836</a>  |
| STAMP      | Kyrouac, J., Cook, D., Ermold, B., Pal, S., Sullivan, R., & Keeler, E. Soil Temperature and Moisture Profiles (STAMP), 2025-01-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Tower Site) (S10). Atmospheric Radiation Measurement (ARM) User Facility. <a href="https://doi.org/10.5439/1238260">https://doi.org/10.5439/1238260</a>                                                                 |